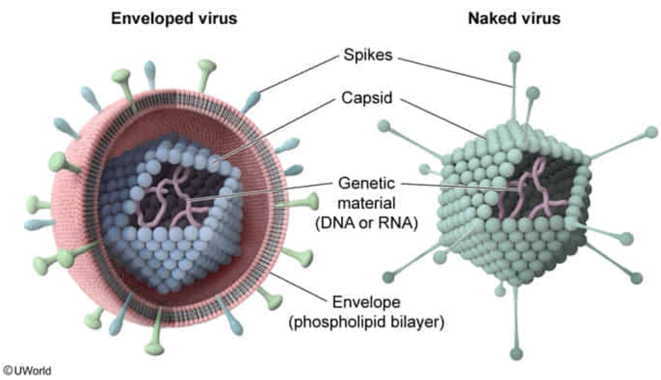
A pathogen identified in a patient's blood sample is found to lack a phospholipid bilayer. Therefore, any medications used to treat the disease caused by this pathogen would most likely be classified as:

- A. antiprotozoal drugs. [9%]
- B. antifungal drugs. [7%]
- C. antiviral drugs. [69%]
- D. antibacterial drugs. [13%]

Correct
69% answered correctly

Explanation:



Unlike prokaryotic and eukaryotic cells, viruses are obligate intracellular parasites that cannot reproduce outside of a host cell. The genetic material of viruses can be RNA or DNA that is double-stranded or single-stranded and is surrounded by a protective protein coat known as the capsid.

In addition, viruses can be either enveloped or nonenveloped. **Enveloped viruses** have a membrane, or phospholipid bilayer, generally derived from the cell membrane of the host. The phospholipid bilayer of enveloped viruses surrounds the capsid and often contains host-derived proteins, allowing the virus to better evade the immune system and gain entry into the host cell. In contrast, **nonenveloped** (naked) viruses **lack a phospholipid bilayer**.

In this scenario, the pathogen identified from the patient's blood sample was found to lack a phospholipid bilayer. Because *all* prokaryotic (eg, bacteria) and eukaryotic (eg, protozoa, fungi) cells are enclosed by a phospholipid bilayer (ie, cell membrane), the only possibility is that the pathogen lacking this bilayer is a virus (nonenveloped); therefore, it should be treated with antiviral drugs.

(Choices A and B) Antifungal drugs are effective against fungi, and antiprotozoal drugs are effective against protozoa. However, protozoa and fungi are classes of eukaryotic cells (*not* viruses), and *all* eukaryotic cells

have a cell membrane. As a result, an organism without a phospholipid bilayer cannot be eukaryotic, making treatment with such drugs ineffective.

(Choice D) Bacteria are prokaryotic cells. Although several factors distinguish prokaryotes from eukaryotes, *all* prokaryotic cells are surrounded by a phospholipid bilayer (cell membrane). Therefore, the pathogen identified in the question cannot be bacterial, and antibacterial drugs would not be effective against it.

Educational objective:
Viruses possess DNA or RNA genomes surrounded by a protein capsid and are unable to reproduce outside of a host. Viruses can also be classified as either enveloped (have a phospholipid bilayer as the cell membrane) or nonenveloped (no phospholipid bilayer). In contrast, all prokaryotic and eukaryotic cells are enclosed by a phospholipid bilayer.

Time spent:0 sec	**QID**:402082
Subject:	**System:**
Biology	Cellular Biology